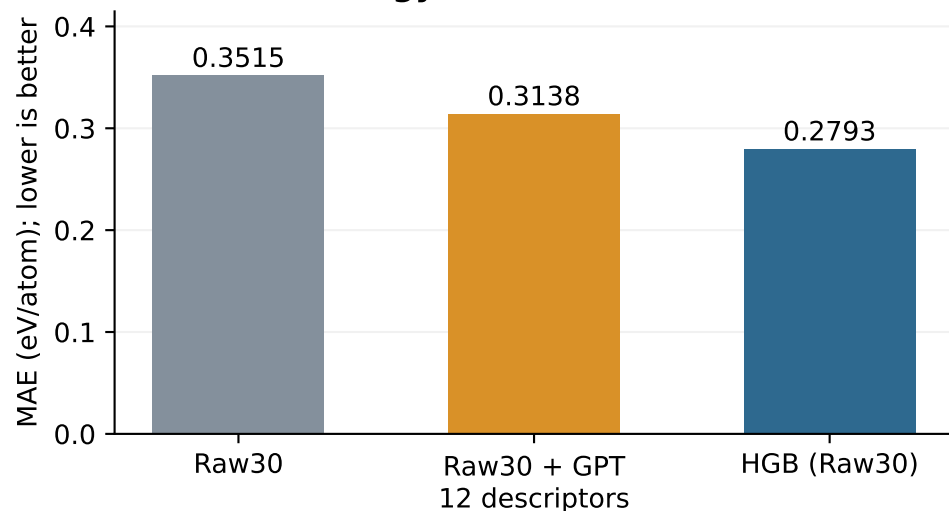


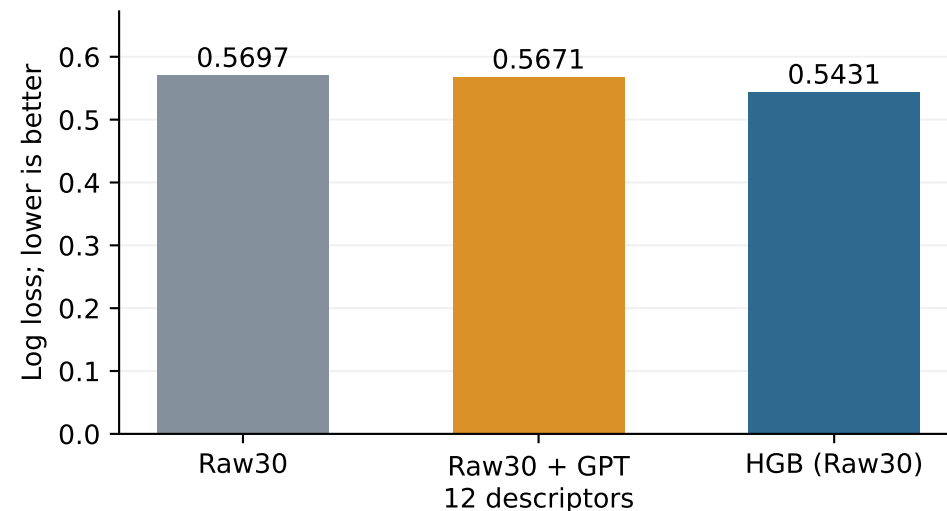
Tool-using GPT descriptor discovery: frozen independent test

721 Materials Project structures · 681 held-out chemical systems · same rows in all arms

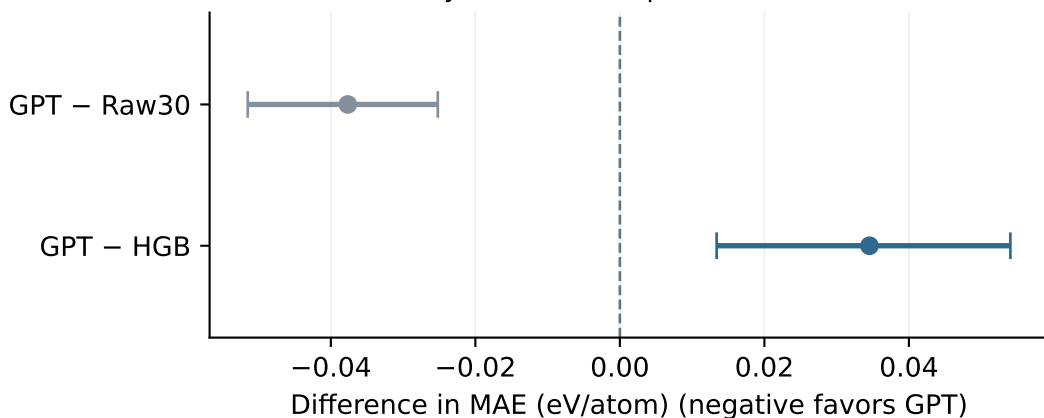
Formation energy



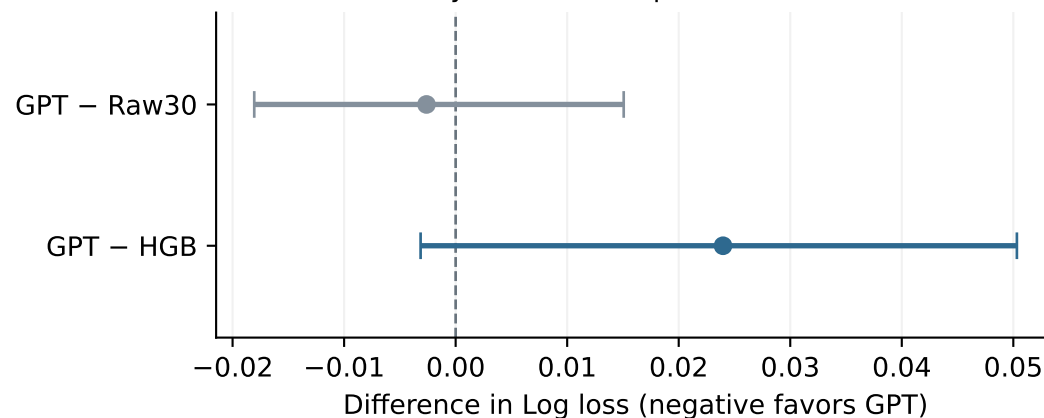
On-hull classification



Paired chemical-system bootstrap: 95% interval



Paired chemical-system bootstrap: 95% interval



Intervals: 1,000 paired system resamples, conditional on the fitted models.

DFT-relaxed inputs; no new DFT, superconductivity test, or matched broad descriptor-search control.